



SECURITY + GOVERNANCE FOUNDATION

Azure Managed Identities · Key Vault · Microsoft Entra ID

Every agent is identity-bound · No secrets in code · No direct table writes · Full lineage from data source through reasoning chain to output · AgentLog captures every decision, citation, and entity verification

SHARED ENTERPRISE CONTEXT

MICROSOFT IQ INTELLIGENCE LAYER · FABCON 2026 · ALL THREE SIMULTANEOUSLY LOAD-BEARING

Shared Enterprise Context one version of reality for every agent in the mesh

Eliminates semantic drift the number one source of hallucinated decisions in multi-agent systems. Remove any one layer and the system fails architecturally.

Foundry IQ

kb-amosx-prod · 10 enterprise policy documents · Answer synthesis · GPT-4.1-mini · Every decision grounded at decision time with exact citations: document name, policy ID, section

Zero hallucinated decisions

Work IQ

Organizational graph surfacing · Work IQ User + Work IQ Mail tools · Exact human routing for every escalation · Backup owner identification during organizational risk detection

Right person. Right moment.

Fabric IQ

12-entity ontology: Order, Customer, Inventory, SLA, Carrier, Employee, Team, Project, Vendor, Contract, Risk, Decision · Semantic contract between all agents · Entity graph traversal

Zero semantic drift

A2A PEER-TO-PEER · IGNITE 2025

AZURE EVENT GRID · EVGNS-AMOSX-PROD-EUS · EAST US · A2A ARCHITECTURE · IGNITE 2025

Decentralized agent mesh spine · No master agent · No central controller · No single point of failure

Each agent subscribes independently and responds based on its own system prompt and Foundry IQ grounding. Implements the A2A peer-to-peer architecture Microsoft introduced at Ignite 2025.

order.customer_inquiry

sla.warning

sla.critical

risk.emerging

escalation.approved

INDEPENDENT REASONING

AZURE AI FOUNDRY · AIF-AMOSX-PROD-EUS · GPT-4.1-MINI · 5 AUTONOMOUS AGENTS · EACH REASONS INDEPENDENTLY OVER THE SHARED ENTERPRISE CONTEXT



Context Agent

ag-amosx-dynamics-prod
context_agent.py

Builds Shared Enterprise Context from Azure SQL and Dynamics 365. Single source of truth for all agents. All data retrieval goes through this agent only.

OpenAPI → Logic App



Reasoning Agent

ag-amosx-sla-prod
sla_monitor.py

Executes 5-step Consequence Cascade Reasoning chain. Foundry IQ citations. Fabric IQ semantic verification. Every decision logged with full citation chain.

GitHub Copilot



Immune Agent

ag-amosx-immune-prod
immune_agent.py

Runs every 15 minutes unprompted. Detects operational and organizational failures before any human reports a problem. "No one asked me to do this. I noticed."

GitHub Copilot



Risk Response Agent

ag-amosx-notify-prod
risk_response_agent.py

Routes P1-P4 escalations to the exact right human via Work IQ. Executes mitigation workflows. Approval routing through NERVA Command Center.

P1-P4 Escalation Matrix



Voice Agent

ag-amosx-voice-prod
voice_agent.py

Handles inbound calls via Voice Live, Ava Dragon HD. Retrieves live order data. Publishes order.customer_inquiry to Event Grid triggering the full mesh cascade.

GitHub Copilot



NERVA Command Center Copilot Studio

Human-in-the-loop governance layer inside Microsoft Teams · P1 escalation approvals · Executive review · Natural language operational queries · getSLARiskOrders · getOrderStatus · getCarrierAnomalies · Work IQ organizational context · The agents coordinate themselves through Event Grid the Command Center governs, not orchestrates

Enterprise Agents Track

SP_AGENTROUTER

INTEGRATION LAYER

Logic App Router (la-amosx-sql-prod) · sp_AgentRouter(@Action, @OrderID) · API Management

Single entry point for all agent SQL operations · No agent has direct database access · Stored procedures only · Routes by @Action parameter · No agent writes directly to any table · Defaults to getOrderStatus when no action provided · API Management handles throttling and rate limiting

DATA LAYER · DUAL SOURCE · DYNAMICS 365-READY

Azure SQL (db-amosx-prod) · Dynamics 365-ready integration layer

Azure SQL Operational Data

Tables (7): Orders · Inventory · Deliveries
SLA · CarrierPerformance · ImmuneAlerts
AgentLog (observability genome)
Procedures (9): sp_AgentRouter
sp_GetOrderStatus · sp_GetSLARiskOrders
sp_GetCarrierAnomalies · sp_ProceduralMemory
sp_RiskResponse

Dynamics 365 Organizational Data

Customer accounts · ARR values
Contact relationships · Account ownership
Case history · Complaint patterns
Contract tiers · SLA agreements
Prepared for Shared Enterprise Context enrichment through Context Agent



AgentLog · Every decision captured with full reasoning record · Agent name · Foundry IQ citation · Fabric IQ entity verified · Confidence score · Event published · Execution time in milliseconds · No direct database access from any agent · All SQL operations route through sp_AgentRouter only

GOVERNANCE + LEARNING LAYER

AgentLog · RiskResponses · Procedural Memory · Organizational Learning

Every decision NERVA has ever made is captured, cited, and available for organizational learning · AgentLog records agent name, decision type, input data, reasoning chain, Foundry IQ citation, Fabric IQ entity verified, confidence score, event published, and execution time in milliseconds · Procedural Memory enables pattern reuse across future incidents · The organization gets smarter with every reasoning execution

INDUSTRY INDEPENDENT

Supply Chain

Healthcare

Manufacturing

Finance

Humanitarian

Any Enterprise

The entities change. The reasoning engine does not.

NERVA does not monitor systems. NERVA monitors consequences.

github.com/Gehanmassoud/amoms-x · Azure AI Foundry · All three Microsoft IQ layers · GitHub Copilot · Contextualize · Anticipate · Intervene · Evolve